

Week 39, Tuesday Session

Flipped classroom: your questions, then how to write the report – abstract, introduction, conclusion, references and the LLM declaration

Bring your laptop and your Project 1 notes – we work in `week39tuesday.ipynb`

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How today works

The format (as every Tuesday)

Mondays are lectures; Tuesdays belong to your questions and to concrete cases. Today the case is not a piece of code but a piece of text: the report you will hand in on October 5.

1. **First ~20 min:** your open questions from Monday's lecture – SGD once more, logistic regression, Adam and standardisation, the neuron – discussed in plenum; then one slide on the tool for the writing, $\text{\LaTeX}$ with Overleaf.
2. **Rest of the double session:** *two* cases, worked in your project groups from `week39tuesday.ipynb`. Case 1: the abstract and the introduction of *your* Project 1 report. Case 2: the conclusion, the reference list, and the appendix that declares how you used large language models. These five pieces are the weekly exercise, due Friday, and you reuse them verbatim in the report.
3. **Last 5 min:** wrap-up, deadlines, and what next Monday builds on.

Picking up from Monday – and why the report matters

Monday in one sentence each

SGD: an epoch costs one full step and buys hundreds; decay γ or rattle. Logistic regression: linear log-odds, cross entropy from the likelihood, gradient $X^T(p - y)$, Newton in a handful of steps. Adam and standardisation: the adaptive methods remove the column scales from the curvature, not from the distance each parameter must travel – standardise first. The neuron: the perceptron with a sigmoid; its learning rule is the delta rule, SGD with $M = 1$. (Automatic differentiation waits for week 40.)

Questions we expect

Why does $\gamma < 8/\lambda_{\max}$ hold at $\theta = 0$ and not at the optimum? If Adam is scale invariant, why is Adam on raw features still seven times worse than plain SGD on standardised ones? Why did the sigmoid replace the perceptron's step function? What does "a penalty is necessary" mean for scikit-learn's defaults? Ask.

How Project 1 is graded (EvaluationForm.md, 100 points)

Abstract: accurate and informative?	5
Introduction: status of the problem, objectives	10
Formalism/methods	10
Code, implementation, testing	10
Analysis of results	40
Conclusions, critical comments, perspectives	10
LLM declaration	5
Overall presentation (figures, tables)	5
Referencing	5

Today's five pieces are worth 35 points before a single number is computed – and they are the parts a tired reader reads first, and last.

The tool: writing the report in $\text{\LaTeX}$ with Overleaf

Why $\text{\LaTeX}$, and why Overleaf

We ask for the report as a scientific article typeset in $\text{\LaTeX}$: equations, numbered figures and tables, cross-references and a bibliography are what it is made for. **Overleaf** (<https://www.overleaf.com>) is $\text{\LaTeX}$ in the browser – nothing to install, the PDF next to the source, and the whole group writes in the *same* document, with history and comments.

```
\section{Results}\label{sec:results}
Figure~\ref{fig:mse} shows the test MSE
for Ridge \cite{hoerl1970}.
\begin{figure}
  \includegraphics[width=\linewidth]
    {Figures/mse_ridge.pdf}
  \caption{Test MSE against  $\lambda$ .}
  \label{fig:mse}
\end{figure}
\bibliography{biblio}
```

Getting started

1. Log in with your UiO e-mail (single sign-on), or make a free account.
2. Upload the course template
FYS_STK_Template.zip
(doc/LectureNotes/data/ in the repository) as a
New Project: main.tex, biblio.bib, Figures/.
3. *Share* the project with your group; send the PDF, not the link, to Canvas.

Habits that save you the last night

Save figures from the code as PDF
(`plt.savefig("mse_ridge.pdf")`) and upload them
– never screenshots. Refer with `\label/\ref` and
`\cite`, never by typed numbers. One .bib file,
checked against the sources (Case 2). The code lives
in your GitHub repository; the report links to it.
Prefer to work offline? TeX Live/MacTeX with
VS Code compiles the same files.

Case 1: the abstract and the introduction of your Project 1 report

Goal: leave with a draft abstract and a draft introduction that you will reuse in the report (notebook, Case 1). Work in your project group; one document, everybody writes.

1. **Read two abstracts** (notebook): the PRL abstract from `projectwriting` and the Project 1 example. Mark, in each, the five moves of the grading guide: *topic and why it matters* – *the open question* – *what was done* – *the main result* – *the implication*.
2. **Write your abstract** (120–180 words, one paragraph, no citations, no equations). Numbers you do not have yet go in as placeholders: “the test MSE is lowest at degree [d]”. Every sentence must survive the question “so what?”.
3. **Write the first and the last paragraph of your introduction.** First: the motivation – why fit a function nobody needs to fit (Runge) with linear models? Last: the structure of the report, one sentence per section. The middle paragraphs (what was done, the models, the methods) are a bullet list for now.
4. **Swap with a neighbouring group** and grade each other’s abstract with the form’s question: *accurate and informative?* Circle every sentence that could be in *any* Project 1 report.

Checkpoints (plenary after ~35 min)

Abstract: five moves present, the main result stated as a *number* or a *comparison*, no “in this report we will”. Introduction: opens with the problem, not with “machine learning is important”; ends with the map of the report; announces what is *yours* (own code, own derivations).

Case 1: what a Project 1 abstract can look like

Example (numbers are placeholders – yours will differ)

Polynomial regression is the simplest setting in which the central questions of supervised learning – model complexity, regularisation and the reliability of an error estimate – can be studied exactly. We fit Runge's function $1/(1 + 25x^2)$ on $[-1, 1]$, sampled at $n = []$ points with Gaussian noise of standard deviation $[],$ with polynomials of degree up to 15 using ordinary least squares, Ridge and Lasso regression, and we compare the closed-form estimators with our own implementations of gradient descent, momentum, AdaGrad, RMSProp, Adam and stochastic gradient descent. Model selection is done by the bootstrap and by k -fold cross-validation, and the test error is decomposed into bias and variance. We find that the test error is minimised at degree $[],$ for OLS and that Ridge with $\lambda = []$ extends the usable range of degrees to $[],$ at the price of a bias of $[],$; Lasso sets $[],$ of the $[],$ coefficients to zero at the cross-validated $\lambda.$ Gradient descent reproduces the closed-form solutions to $[],$ within $[],$ iterations only after the features are standardised, and Adam is the most robust to the choice of learning rate. Our results illustrate that resampling, not the training error, decides the model, and that the optimiser and the preprocessing are part of the model.

Why it works

Move 1 (why it matters) is one sentence. Move 2 (the question) is implicit in “compare”. Move 3 says exactly what was done, including *own* implementations. Move 4 gives numbers. Move 5 is one sentence and is not a platitude. 175 words, past tense for what was done, present tense for what is true.

Case 1: the introduction, paragraph by paragraph

1. Motivation (the reader's problem, not yours)

Why does anyone fit functions with polynomials? Why is Runge's function the classic test? What goes wrong at high degree, and what is the general lesson (bias-variance, Section 2.10)? Two to three paragraphs; cite the sources you actually used.

2. What has been done, and what you do

The methods in one paragraph each: the linear models (Chapter 3), the resampling methods (Sections 2.11, 3.14), the optimisers (Chapter 4). State what is *your* contribution: own code, own derivations, own experiments – and what is a library call (scikit-learn for cross-validation is fine; say so).

3. The structure of the report (always the last paragraph)

“Section 2 presents the models and the bias-variance decomposition; Section 3 describes our implementation and the tests against the closed-form solutions; Section 4 presents the results, first for OLS, then Ridge and Lasso, then the optimisers; Section 5 concludes. The code is available at [URL].”

What does not belong here – and the voice

Results (they come in Section 4 – but the abstract may quote them). Long derivations. Definitions of MSE and R^2 (Methods). “In recent years, machine learning has become. . .” Voice: “we” throughout, past tense for what you did, present tense for what is true. Written *last*, read *first*.

Case 1: do you have the insight?

1. A group writes “In this project we will implement OLS, Ridge and Lasso and study the bias–variance trade-off” as their abstract. Which of the five moves are present, which are missing, and why is the tense wrong?
2. Your test MSE for OLS is lowest at degree 7 with the value 0.0121 (noise variance 0.01). Does this number belong in the abstract, in the introduction, in both, or in neither?
3. The introduction’s last paragraph lists the sections. Why do we insist on it, when the table of contents already does that?
4. You used `scikit-learn` for cross-validation and your own code for everything else. Where in the introduction does the reader learn this, and why does it matter for the grade?

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What we hope you say

(1) Only move 3, and only as a plan: no motivation, no question, no result, no implication – and “we will” describes work not yet done; an abstract reports work that *was* done. (2) In the abstract, yes – it is the main result and a reader deciding whether to read on wants it. In the introduction, no: the introduction says *that* we compare degrees and *how*; the number belongs to Results. (3) The table of contents lists headings; the last paragraph tells the reader what the *argument* is and where each step of it lives – it is the only place where the logic of the report is stated in one breath. (4) In the paragraph on what was done, explicitly: “own implementation of ...; cross-validation with `scikit-learn`”. The form grades code, implementation and testing (10 points) and the analysis (40); the grader needs to know what to credit you for, and concealing a library call reads as a claim you did not earn.

Case 2: the conclusion, the references, and the declaration of LLM use

Goal: a draft conclusion, a reference list in the required format, and the LLM appendix filled in for what you have actually done so far (notebook, Case 2).

1. **The conclusion, three paragraphs:** (i) main findings and their interpretation – the abstract’s numbers, now *explained*; (ii) the critical paragraph: what the methods could and could not do, what you would do differently, where your results are least certain; (iii) perspectives – the one or two things you would do next, and why. Write (ii) first; it is the hardest and the most valuable.
2. **References:** list every source you have *used* – the book chapters, Hastie et al., Goodfellow et al., the papers behind Ridge, Lasso and Adam, the software you depend on. Format: articles as *authors, journal, volume (bold), page (year)*; books as *authors, title, publisher, place, year*. Set them up as a .bib file (notebook) and cite each one at least once.
3. **Verify every reference** against the actual source – not against another LLM. Two of the references in the notebook are wrong on purpose. Find them.
4. **The LLM appendix:** tools used; one contribution level (0–3) per section of the report; one level (0–4) per code file; the LLM-assisted docstring tag on every function at level 2 or above. Fill it in *now* for the abstract and introduction you just wrote, and for the code you have.

Checkpoints (plenary after ~30 min)

Conclusion: no sentence repeats the abstract verbatim; paragraph (ii) exists; perspectives are concrete (“Lasso path with warm starts”, not “more research is needed”). References: correct format, verified, cited. LLM appendix: present, specific, honest – and *consistent with the docstrings in the code*.

Case 2: a conclusion is not a second abstract (illustrative numbers)

The abstract says

“... the test error is minimised at degree 7 for OLS and Ridge with $\lambda = 10^{-3}$ extends the usable range of degrees to 12 at the price of a bias of 0.002...”

The conclusion explains

“At degree 7 the bias of the OLS fit (0.003) and its variance (0.008) are of the same size, and both bootstrap and 10-fold cross-validation locate the minimum there; beyond degree 9 the variance grows by a factor of [] per degree while the bias no longer decreases. Ridge damps the variance in the directions of the small singular values of X (Section 3.8), which is why it extends the usable degree; the bias it introduces is visible as the systematic underestimate of the peak of f . Lasso’s zeros coincide with the odd powers, as the symmetry of f requires – a check that the penalty selects, not merely shrinks.”

The critical paragraph

“Our bias–variance estimates use the known f ; on real data the bias term would absorb the noise variance, and our numbers are optimistic by σ^2 . The gradient-descent comparisons were run at a fixed number of iterations rather than to a tolerance, which favours Adam; and the learning rates were scanned on a grid of 16 values, coarse enough that the ranking of momentum and RMSProp at degree 10 is not resolved.”

Perspectives, concretely

“A natural next step is the Lasso path with warm starts along λ , which would replace the [] separate fits by one; and a validation-based stopping rule for SGD, which would remove the iteration count as a hyperparameter. Both are needed before the same pipeline is applied to the classification data of Project 2.”

Case 2: references – what to cite, and how

Cite what you used: the method's origin, the text you learned it from, the software you depend on, the data.

Not: things you did not read.

```
@book{hastie2009,  
  author = {Hastie, Trevor and Tibshirani, Robert and Friedman, Jerome},  
  title = {The Elements of Statistical Learning},  
  edition = {2},  
  publisher = {Springer},  
  address = {New York},  
  year = {2009}  
}  
@article{tibshirani1996,  
  author = {Tibshirani, Robert},  
  title = {Regression shrinkage and selection via the lasso},  
  journal = {Journal of the Royal Statistical Society B},  
  volume = {58},  
  pages = {267--288},  
  year = {1996}  
}  
@article{pedregosa2011,  
  author = {Pedregosa, Fabian and others},  
  title = {Scikit-learn: Machine Learning in {P}ython},  
  journal = {Journal of Machine Learning Research},  
  volume = {12},  
  pages = {2825--2830},  
  year = {2011}  
}
```

The course format (projectwriting)

Articles: name(s) of author(s), journal, **volume**, page (year).

Books: name(s) of author(s), title, publisher, place, year.

With revtex (latexexample.tex) this is `\bibliographystyle{apsrev4-1}`; with article, unsrt or plain does it. Cite with `\cite{hastie2009}`; a reference nobody cites is deleted by BibTeX – and by the grader.

Verify – always

Open the source. Check authors, year, volume, pages, and that it says what you claim. Language models invent plausible references with real authors and wrong titles, and confidently: “Tibshirani, R. (1994), *The Lasso method*, Ann. Statist.” does not exist. An unverified reference in your report is *your* error.

Case 2: declaring the use of large language models

The principles (the guidelines file in the repository)

LLMs are *allowed and encouraged*. Using one does not reduce your responsibility: every equation, number, code block and sentence must be something you understand, can defend, and have verified. Declare *significant* use (content, logic or structure that appears in the submission) – not a tutoring conversation you summarised yourself. Concealment, not use, is the integrity violation. 5 points of the grade.

Where

An appendix “Use of AI/LLM Tools”: A.1 tools (name, version/date, access); A.2 one contribution level per *section*; A.3 one level per *code file*; plus the LLM-assisted docstring tag in the code and a markdown note above assisted notebook cells. Level 3 text and level 3–4 code get a sentence on the prompt and on what you changed.

Text levels: 0 none; 1 language only; 2 editorial (rewrote your sentences); 3 generative (drafted from your prompt; you revised and own it).

Code levels: 0 none; 1 debugging; 2 snippet; 3 skeleton; 4 substantial.

```
\section*{Appendix A: Use of AI/LLM tools}
\textbf{A.1 Tools used.} Claude (claude.ai, Sept--Oct 2026);
GitHub Copilot (VS Code, throughout).
\textbf{A.2 Text.}
\begin{tabular}{llp{5cm}}
Section & Level & Notes\\ \hline
Abstract & 1 & grammar check\\
Introduction & 2 & two paragraphs rephrased for clarity\\
Methods & 0 & --\\
Results & 0 & --\\
Conclusion & 1 & minor language corrections\\
\end{tabular}
\textbf{A.3 Code.}
\begin{tabular}{llp{5cm}}
File & Level & Description\\ \hline
regression.py & 0 & written independently\\
optimisers.py & 2 & Copilot suggested the Adam bias correction lines
;
; tested against the closed-form OLS solution\\
plots.ipynb & 1 & Claude found an axis-indexing bug\\
\end{tabular}
```

Case 2: do you have the insight?

1. You asked an LLM to explain the bias–variance decomposition, then wrote Section 2.3 yourself from your own notes. Level? And if you pasted its derivation and fixed the notation?
2. Copilot autocompleted most of your plotting code line by line. Does every accepted line need a comment? What goes in the appendix?
3. A reference in your list has the right authors, the right journal and the wrong year and pages. Whose error is it, and how would the grader find it?
4. Your conclusion's critical paragraph says the learning-rate grid was too coarse to rank two optimisers. Does admitting this lower or raise your score under "Conclusions, discussions and critical comments"?

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What we hope you say

(1) Level 0 – an LLM used as an interactive textbook, with the writing entirely yours, is not declared (guidelines, FAQ 1). Pasting and fixing the notation is level 3: declare it, give the prompt, say what you changed, and be able to derive it at the oral follow-up. (2) No – boilerplate autocompletion is declared once, at file level, “Copilot used throughout for autocompletion”; any multi-line, logically non-trivial suggestion you accepted gets a file-level note and, if it is a function, the docstring tag. (3) Yours. Every citation is a claim you make; the grader opens the DOI or the journal page – “relevant works cited *accurately*?” is the criterion, and a wrong year and pages are what an invented reference looks like. (4) Raises it. The criterion asks for critical comments on what was learned about the method and the results; a report that knows the limits of its own numbers is worth more than one that does not. What lowers the score is a limitation the grader finds and you did not mention.

Wrap-up, deadlines, and Monday

- ▶ **What you should own now:** a draft abstract with the five moves and placeholders for numbers; the first and last paragraph of the introduction; a three-paragraph conclusion with a critical paragraph; a verified .bib file in the course format; and the LLM appendix filled in for the work done so far, consistent with the docstrings in your code.
- ▶ **Weekly exercises** (exercisesweek39.ipynb): exactly these five pieces, written for your own Project 1. Deadline **Friday September 25 at midnight**. You get short feedback before the project deadline, and you reuse the pieces in the report.
- ▶ **Project 1** is due **October 5**. Parts a)–i) are done in the Tuesday cases of weeks 35–38; what remains is the analysis, the text – and running the whole pipeline once more from a clean start to make sure the numbers in the report are the numbers the code produces.
- ▶ **Next Monday (week 40):** neural networks (Chapter 8) – the neuron of this Monday, in layers, and the part we left out: the same picture read backwards, automatic differentiation (Section 5.8.2), which with one layer more is backpropagation.

Before you leave

One line in an email to us: the muddiest point of the week – what should Monday's first ten minutes revisit?